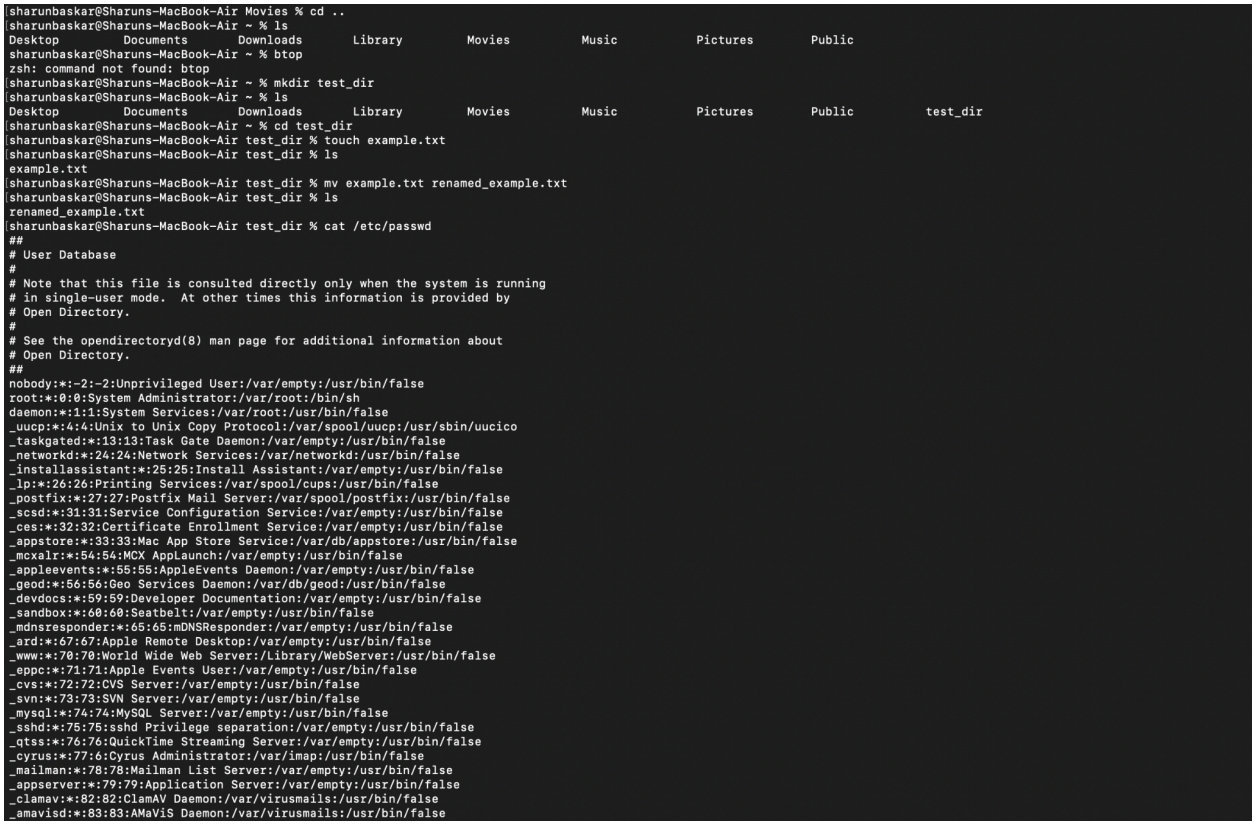
# Linux Basics Assignment




This above SS contains the workout for the below Tasks.

## 1. Creating and Renaming Files/Directories

1. Create a directory named test_dir using `mkdir`.
2. Inside test_dir, create an empty file called example.txt.
3. Rename example.txt to renamed_example.txt using `mv`.

## 2. Viewing File Contents

4. Use `cat` to display the contents of /etc/passwd.

```
sharunbaskar@Sharuns-MacBook-Air test_dir % head -n 5 /etc/passwd

##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode.  At other times this information is provided by
sharunbaskar@Sharuns-MacBook-Air test_dir % tail -n 5 /etc/passwd

_audiomxd:*:294:294:Audio and MediaExperience Daemon:/var/db/audiomxd:/usr/bin/false
_terminusd:*:295:295:Terminus:/var/empty:/usr/bin/false
_neuralengine:*:296:296:AppleNeuralEngine:/var/db/neuralengine:/usr/bin/false
_eligibilityd:*:297:297:OS Eligibility Daemon:/var/db/eligibilityd:/usr/bin/false
_oahd:*:441:441:OAH Daemon:/var/empty:/usr/bin/false
sharunbaskar@Sharuns-MacBook-Air test_dir %
```

5. Display only the first 5 lines of /etc/passwd using `head`.
6. Display only the last 5 lines of /etc/passwd using `tail`.

```
sharunbaskar@Sharuns-MacBook-Air test_dir % grep "root" /etc/passwd

root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_cvmsroot:*:212:212:CVMS Root:/var/empty:/usr/bin/false
sharunbaskar@Sharuns-MacBook-Air test_dir % zip -r test_dir.zip test_dir
        zip warning: name not matched: test_dir

zip error: Nothing to do! (try: zip -r test_dir.zip . -i test_dir)
sharunbaskar@Sharuns-MacBook-Air test_dir % cd ..
sharunbaskar@Sharuns-MacBook-Air ~ % zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
sharunbaskar@Sharuns-MacBook-Air ~ % ls
Desktop         Documents       Downloads       Library         Movies          Music           Pictures        Public          test_dir        test_dir.zip
sharunbaskar@Sharuns-MacBook-Air ~ % unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
   creating: unzipped_dir/test_dir/
 extracting: unzipped_dir/test_dir/renamed_example.txt
sharunbaskar@Sharuns-MacBook-Air ~ % ls
Desktop         Documents       Downloads       Library         Movies          Music           Pictures        Public          test_dir        test_dir.zip    unzipped_dir
sharunbaskar@Sharuns-MacBook-Air ~ % cd unzipped_dir
sharunbaskar@Sharuns-MacBook-Air unzipped_dir % ls
test_dir
```

### 3. Searching for Patterns

7. Use `grep` to find all lines containing the word “root” in /etc/passwd.

### 4. Zipping and Unzipping

8. Compress the test_dir directory into a file named test_dir.zip using `zip`.
9. Unzip test_dir.zip into a new directory named `unzipped_dir`.

```
sharunbaskar@Sharuns-MacBook-Air ~ % wget https://example.com/sample.txt
--2026-09-10 20:35:35--  https://example.com/sample.txt
Resolving example.com (example.com)... 2606:4700:83b5:72db:f2fd:50c:ef6b:ff98, 172.66.147.243, 104.20.23.154
Connecting to example.com (example.com)|2606:4700:83b5:72db:f2fd:50c:ef6b:ff98|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-09-10 20:35:35 ERROR 404: Not Found.

sharunbaskar@Sharuns-MacBook-Air ~ % s
zsh: command not found: s
sharunbaskar@Sharuns-MacBook-Air ~ % ls
Desktop         Library         Pictures        test_dir.zip
Documents       Movies          Public          unzipped_dir
Downloads       Music           test_dir
sharunbaskar@Sharuns-MacBook-Air ~ % touch secure.txt
sharunbaskar@Sharuns-MacBook-Air ~ % ls
Desktop         Library         Pictures        test_dir
Documents       Movies          Public          test_dir.zip
Downloads       Music           secure.txt      unzipped_dir
sharunbaskar@Sharuns-MacBook-Air ~ % chmod 444 secure.txt
sharunbaskar@Sharuns-MacBook-Air ~ % ls
Desktop         Library         Pictures        test_dir
Documents       Movies          Public          test_dir.zip
Downloads       Music           secure.txt      unzipped_dir
sharunbaskar@Sharuns-MacBook-Air ~ % ls -l
total 8
drwx------+  7 sharunbaskar  staff   224 Sep 10 20:36 Desktop
drwx------+  3 sharunbaskar  staff    96 Sep  3 09:58 Documents
drwx------+  5 sharunbaskar  staff   160 Sep 10 09:59 Downloads
drwx------@ 84 sharunbaskar  staff  2688 Sep  3 10:25 Library
drwx------   3 sharunbaskar  staff    96 Sep  3 09:58 Movies
drwx------+  4 sharunbaskar  staff   128 Sep  4 10:05 Music
drwx------+  5 sharunbaskar  staff   160 Sep  3 12:28 Pictures
drwxr-xr-x+  4 sharunbaskar  staff   128 Sep  3 09:58 Public
-r--r--r--   1 sharunbaskar  staff     0 Sep 10 20:38 secure.txt
drwxr-xr-x   3 sharunbaskar  staff    96 Sep 10 09:13 test_dir
-rw-r--r--   1 sharunbaskar  staff   352 Sep 10 20:16 test_dir.zip
drwxr-xr-x   3 sharunbaskar  staff    96 Sep 10 20:18 unzipped_dir
sharunbaskar@Sharuns-MacBook-Air ~ % export MY_VAR="Hello, Linux!"
dquote>
dquote>
dquote>
dquote> echo $MY_VAR
dquote>
sharunbaskar@Sharuns-MacBook-Air ~ % export MY_VAR='Hello, Linux!'
sharunbaskar@Sharuns-MacBook-Air ~ % echo "$MY_VAR"
Hello, Linux!
sharunbaskar@Sharuns-MacBook-Air ~ %
```

## 5. Downloading Files

10. Use `wget` to download a file from a URL (e.g., https://example.com/sample.txt).

## 6. Changing Permissions

11. Create a file named secure.txt and change its permissions to read-only for everyone using `chmod`.

## 7. Working with Environment Variables

12. Use `export` to set a new environment variable called MY_VAR with the value “Hello, Linux!”.